MLET: Incremental Predictive Value of OpenET and an Auditable Reference-Evapotranspiration Outlook

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Abstract

Machine Learning Evapotranspiration (MLET) separates two targets. Reference evapotranspiration (ETo, denoted ET_o below) describes atmospheric demand for a defined reference surface. Actual evapotranspiration (ET_a) also depends on crop condition, soil water, and management. MLET therefore keeps two evidence paths. The retrospective path tests OpenET for daily actual-ET prediction. The prospective path defines an auditable 20-day reference-ETo artifact for Idaho. The reproduced Phase 2 record contains 7,923 station-days from 85 stations. The preregistered comparison is M3 OpenETRidge against B2 WeatherRidge. M3 has mean absolute error (MAE) 0.856 mm day^{-1} versus 1.514 mm day^{-1} for B2. This is a 0.658 mm day^{-1} improvement and a 43.4% reduction. A 2,000-replicate station-blocked bootstrap gives a 95% interval of [0.399, 0.911] mm day^{-1}. M2 OpenETRecal has the lowest descriptive MAE, 0.781 mm day^{-1}, but it is not the preregistered arm. The prospective path applies the ASCE standardized short-reference equation to archived ensemble members. One real issue, one station, and 20 targets pass source and identity checks. That retrospective reforecast diagnostic has MAE 1.133 mm day^{-1}, while prior-station climatology has MAE 0.505 mm day^{-1}. The signed baseline-minus-forecast MAE difference is -0.628 mm day^{-1}, so forecast MAE is higher. Empirical p10-to-p90 coverage is 0.25 against a nominal coverage target of 0.80, and mean band width is 1.453 mm day^{-1}. One bootstrap cluster and fewer than 30 targets per support cell prevent a confidence interval and a skill claim.

Keywords: evapotranspiration; actual ET; reference ET_o; OpenET; GEFS; AgriMet; forecast audit; provenance; uncertainty

1 Introduction

Evapotranspiration connects the land surface, the atmosphere, and water management. Reference ET_o measures atmospheric demand over a defined reference surface (Allen et al., 1998; ASCE Task Committee, 2005). Actual ET_a is the water flux from the observed crop and soil system. It responds to water stress, crop development, soil condition, and management. A retrospective actual-ET estimate and a prospective reference-ET_o outlook can therefore be individually valid and jointly misleading when they share one label.

Satellite ET products provide broad retrospective information. OpenET joins several remote-sensing models and has been evaluated against in situ ET observations across the contiguous United States (Volk et al., 2023). Weather ensembles provide a different evidence source. They can drive a standardized reference-ET_o equation for future dates, but they do not observe future crop or soil state. The scientific question is not only whether a model has a low error. The question is also whether the target, issue time, baseline, and claim remain unchanged from data acquisition to publication.

This paper reports one completed comparison and one incomplete outlook evaluation. The completed comparison asks whether an OpenET feature improves retrospective daily actual-ET prediction under station-held-out 10-fold evaluation. The outlook protocol asks whether archived Global Ensemble Forecast System version 12 (GEFSv12) weather from the National Oceanic and Atmospheric Administration (NOAA) can support a 20-day reference-ET_o artifact against an independent station-derived target. The protocol fails closed when source time, station identity, or evaluation support is incomplete. A frozen protocol is fixed before evaluation. Promotion means public operational release. Validation complete means that every required support cell meets its target count. A skillful result has the preregistered MAE reduction and a confidence interval above zero. Release review is an independent check before any public release.

The work makes four contributions. First, it defines a target ontology that prevents actual ET, reference ET_o, and conditional crop ET from becoming interchangeable. Second, it reproduces a checksum-bound Phase 2 comparison and reports its preregistered model arm. Third, it implements an issue-time-valid reference-ET_o artifact with quantiles, source receipts, and station-to-grid identity. Fourth, it exposes an incomplete evaluation as a machine-readable state. This last contribution prevents one feasible case from becoming a validation claim.

2 Evidence architecture

Figure 1 gives the governing separation. The left path has historical OpenET values and historical flux labels. The right path has a forecast issue, future local dates, and later AgriMet observations. Results do not cross this boundary.

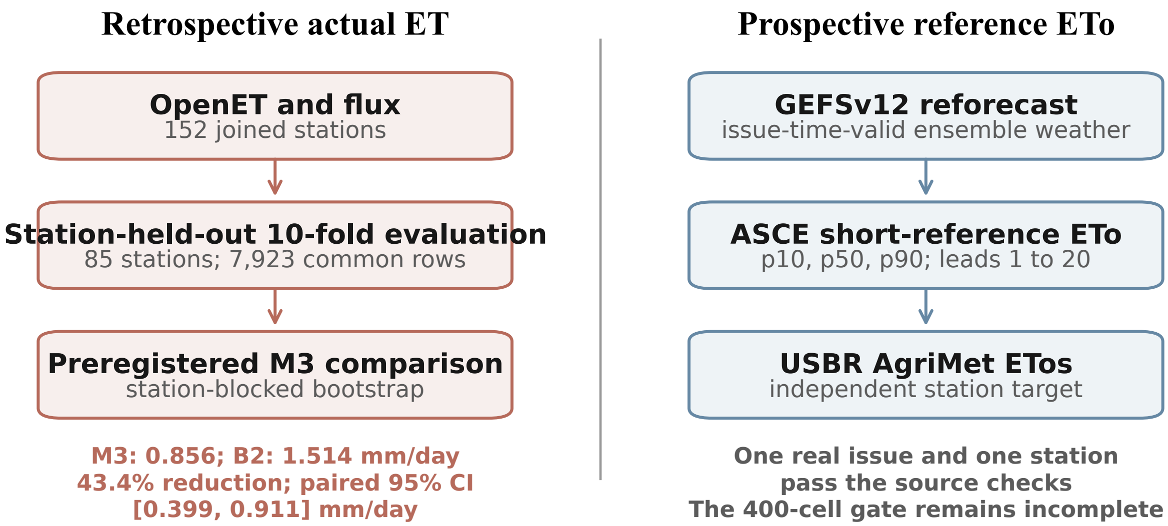

Figure 1. Two targets require two evidence paths. The left result is a retrospective actual-ET comparison. Its fitted-model sample has $n = 7,923$ station-days from 85 stations. The right path is a prospective reference-ET$_o$ protocol. Its one real case supplies $n = 20$ targets, but the complete 400-cell outcome gate remains open.

2.1 Target ontology

Table 1 states the allowed claims. There is no generic future actual-ET layer in the current product contract. The evaluated future layer is `eto_mm`. Every other future ET quantity needs a separate target and a separate protocol.

Table 1. Target ontology and claim boundary.

Layer or label	Quantity	Time and source	Allowed claim
`ET_corr`	Energy-balance-corrected actual ET	Historical flux label	Retrospective daily actual-ET prediction
`eto_mm`	ASCE short-reference ET$_o$	Future local date from issue-time GEFS	Regional reference-ET$_o$ outlook after the frozen hindcast gate
AgriMet `ETos`	Published grass-reference ET	Later station observation; converted from inches per day	Independent reference-ET$_o$ evaluation target
`potential_et_c_mm`	K_cET$_o$ under ample-water assumptions	Conditional future projection	Conditional crop ET only
`eta_analysis_mm`	Satellite actual-ET analysis	Delayed historical observation	Analysis comparison only
Conditional ET$_a$ scenarios	Named water and soil assumptions	Future scenario	Scenario evidence only

2.2 Proof obligations

Each public claim must bind six identities: the physical target, the source artifact, the forecast issue, the valid date, the independent target, and the evaluation split. A missing identity is not imputed. The evaluator returns an incomplete state. The state can contain diagnostic metrics, but it cannot contain a promotion decision.

3 Data and evidence artifacts

3.1 Retrospective actual-ET data

The Phase 2 build joins daily OpenET model ET with a public flux-tower ET archive on exact station and date keys. The source manifest binds the two archives by identifiers and checksums. The joined dataset has 16,447 rows from 152 stations. It has 16,444 non-gap labels. The response y_i is the energy-balance-corrected daily actual ET in mm day^{-1}. The uncorrected `ET` column is never scored. The benchmark reference ETo value E_i^g is gridMET ETo. It is not GEFS ETo. The OpenET input x_i^{OpenET} is the available daily OpenET actual-ET estimate in mm day^{-1}.

The ridge-model comparison uses the common finite weather subset. This subset has 7,923 station-days from 85 stations. Negative measured `ET_corr` values remain in the label set. Negative satellite ET and negative reference ET fail input validation. This asymmetric rule keeps observations intact and rejects physically invalid predictors.

The OpenET source data and the flux benchmark follow the Phase II accuracy study (Volk et al., 2023). The present experiment does not re-estimate that study's site-level accuracy. It asks a narrower prediction question under one repository-frozen split.

3.2 Prospective weather data

The outlook uses the National Oceanic and Atmospheric Administration (NOAA) Global Ensemble Forecast System version 12 (GEFSv12) weekly reforecast archive (National Centers for Environmental Prediction, 2021). Coordinated Universal Time (UTC) is the time basis for source receipts. A 00Z (00:00 UTC) issue starts at midnight UTC. The frozen schedule contains the Wednesday 00Z issues from 2013-01-02 through 2019-12-25. This is 365 planned issues. The feasibility receipt covers one issue, 2019-07-03 00Z. It finds 187 of 187 planned objects and 8,289,206,079 advertised bytes. These are exact inventory values for one issue. They are not a transfer-performance benchmark.

The decoder produces 42,900 weather rows from 195 points in the common 0.5-degree GEFS grid-point subset, 11 ensemble members, and 20 valid dates. This subset is the exact intersection of the 0.25-degree early source segment and the 0.5-degree late source segment. No interpolation creates it. The identity $195 \times 11 \times 20 = 42,900$ is checked before the ET_o builder runs. Each normalized artifact contains its raw-object receipt and SHA-256 (a 256-bit secure hash) digest.

3.3 Independent AgriMet targets

The U.S. Bureau of Reclamation (USBR) AgriMet network publishes daily grass-reference evapotranspiration (ETos) as reference evapotranspiration (ETo). The American Society of Civil Engineers Environmental and Water Resources Institute (ASCE-EWRI) standard defines this grass reference (U.S. Bureau of Reclamation, 2024a,b). ETos is a published station-derived target. It is not a direct flux measurement and it is not a GEFS recomputation. The acquisition adapter parses the official data boundary. It preserves missing markers as exclusions. It converts each non-missing value from inches per day with the exact factor 25.4.

The current station snapshot has 265 stations, including 52 in Idaho. This snapshot does not establish historical coordinates. Historical page evidence supports 19 stations. Their acquisition record contains 31,406 daily values and 115 exclusions from 2015-06-17 through 2019-12-31. The maximum observed station-to-grid distance is 25.475 km across these 19 mappings. The frozen limit is 50 km. This mapping is station identity, not field geometry.

4 Methods

4.1 Phase 2 station-held-out evaluation

The primary station-held-out 10-fold evaluation assigns stations to $k = 10$ deterministic folds with seed 20260713. A fitted model never sees a row from its held-out station. Feature means, feature scales, coefficients, and the crop coefficient use training rows only. A separate split trains before 2019-01-01 and describes later rows. That time split is not the primary decision split.

The experiment fixes six models. B0 persistence uses the previous labeled day at one station. Its 1,555 consecutive-day pairs make it an oracle-like diagnostic. B0 is separate from the common-sample comparison and from H2. B1 uses one static pooled training-set mean of y_i/E_i^g over rows with $E_i^g > 0$:

$$\widehat{K}_c = \frac{1}{|\mathcal{T}_+|} \sum_{i \in \mathcal{T}_+} \frac{y_i}{E_i^g}, \qquad \mathcal{T}_+ = \{i \in \mathcal{T} : E_i^g > 0\}, \qquad \widehat{y}_i = \widehat{K}_c E_i^g. \tag{1}$$

B1 is not crop-specific or stage-specific. B2 is ridge regression on gridMET ETo, mean temperature, vapor-pressure deficit, wind speed, and sine and cosine day-of-year terms. M1 uses x_i^{OpenET} directly. M2 is ordinary least squares with an intercept and slope:

$$\widehat{y}_i = \widehat{\alpha} + \widehat{\beta} x_i^{\text{OpenET}}. \tag{2}$$

M3 adds OpenET to all B2 predictors. B2 and M3 standardize every predictor with its training mean and standard deviation, then solve

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i \in \mathcal{T}} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right\}, \qquad \lambda = 1. \tag{3}$$

The ridge intercept is the training response mean and is not penalized.

4.2 Deterministic metrics and H2

For prediction $\widehat{y}_i$ and observed target y_i, define the residual $e_{m,i} = \widehat{y}_{m,i} - y_i$ for model m. The experiment reports mean absolute error (MAE), root mean square error (RMSE), and signed bias:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |e_{m,i}|, \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_{m,i}^2}, \tag{5}$$

$$\mathrm{Bias} = \frac{1}{n}\sum_{i=1}^{n} e_{m,i}. \tag{6}$$

H2 (the preregistered OpenET comparison) is a comparison, not a model. It requires at least 10% lower pooled MAE for M3 than the better of B1 and B2, with a station-blocked 95% confidence interval for the baseline-minus-M3 MAE difference above zero. B2 wins the OpenET-free selection. M2 has the lowest MAE only among B1, B2, and M1 through M3 on the common $n = 7,923$ sample. The paired station-day contrast is

$$\Delta_{\mathrm{MAE}} = \frac{1}{n}\sum_{i=1}^{n} \left(|e_{B2,i}| - |e_{M3,i}|\right), \qquad r = \frac{\Delta_{\mathrm{MAE}}}{\mathrm{MAE}_{B2}}. \tag{7}$$

Inference resamples stations, not rows. It uses 2,000 replicates and seed 20260713. The reported interval is the percentile interval of the paired baseline-minus-M3 MAE difference (Efron and Tibshirani, 1993).

4.3 Weather conversion and reference ETo

The GEFS decoder converts each member to local daily ASCE inputs. Air temperature uses $T_C = T_K - 273.15$. Stored GEFS wind is measured at 10 m. The decoder forms the vector magnitude $u_{10} = \sqrt{u^2 + v^2}$, and `pyfao56` performs the internal 10 m to 2 m adjustment. Vapor pressure follows

$$e_a = \frac{qp}{\epsilon + (1-\epsilon)q}\frac{1}{1000}, \qquad \epsilon = 0.622, \tag{8}$$

where q is specific humidity in $\mathrm{kg\,kg^{-1}}$, p is pressure in Pa, and e_a is actual vapor pressure in kPa. Radiation changes from $\mathrm{J\,m^{-2}}$ to $\mathrm{MJ\,m^{-2}}$. Interval midpoints assign weather intervals to the America/Boise civil day. A UTC-day aggregate fails schema validation.

For each member, the code calls the ASCE short-reference equation (ASCE Task Committee, 2005; Thorp, 2022):

$$\mathrm{ET_o} = \frac{0.408\Delta(R_n - G) + \gamma\frac{900}{T+273}u_2(e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)}. \tag{9}$$

Here $\mathrm{ET_o}$ has units $\mathrm{mm\,day^{-1}}$. The symbols are defined as follows. T is mean air temperature (degrees C). R_n is net radiation and G is soil heat flux, both in $\mathrm{MJ\,m^{-2}\,day^{-1}}$. Δ is the vapor-pressure slope and γ is the psychrometric constant, both in $\mathrm{kPa\,degree\,C^{-1}}$. u_2 is 2 m wind speed in $\mathrm{m\,s^{-1}}$. e_s is saturation vapor pressure and e_a is actual vapor pressure, both in kPa. The implementation calls the vendored pyfao56 ASCE daily routine. The vendored software version is 1.4.3.

Let $E_{g\ell j}$ be Eq. 9 for grid point g, lead ℓ, and member j. The artifact stores

$$q_{g\ell}(\alpha) = Q_\alpha\left(\{E_{g\ell j}\}_{j=1}^{m}\right), \qquad \alpha \in \{0.10, 0.50, 0.90\}. \tag{10}$$

The feasibility case has $m = 11$. For each lead, the m values are sorted. The empirical p10, p50, and p90 use NumPy linear interpolation between adjacent order statistics at probabilities 0.10, 0.50, and 0.90. The p10 to p90 result is an uncalibrated ensemble quantile band. The contract rejects duplicate members, missing leads, negative values, and unordered quantiles.

4.4 Issue time and independent target

For issue t_0 and lead ℓ, the valid date is

$$d(t_0, \ell) = \mathrm{date}_{\mathrm{America/Boise}}(t_0) + \ell, \qquad \ell \in \{1, \ldots, 20\}. \tag{11}$$

Source metadata must show historical availability at or before t_0. The target must become available only after its completed local day. Retrieval at a later archive date does not change either information boundary.

For station s and date d, the independent target is

$$y_{sd} = 25.4\ \mathrm{ETos}_{sd}^{\mathrm{in/day}}. \tag{12}$$

The code does not recompute this target from GEFS. It does not replace a missing ETos value with zero or interpolation.

4.5 Leakage-safe climatology

The fixed climatology uses all strictly prior years for the same station and day of year. It is not a learned cross-fold component. For issue year Y_0 and day of year k,

$$b_{s,k,Y_0} = \frac{1}{|\mathcal{Y}_{s,k,Y_0}|}\sum_{Y\in\mathcal{Y}_{s,k,Y_0}} y_{s,k,Y}, \quad \mathcal{Y}_{s,k,Y_0} = \{Y : Y < Y_0\}. \tag{13}$$

The evaluated year and all future years are absent. A missing prior-year set creates an explicit exclusion.

4.6 Probabilistic metrics and support

The outcome protocol reports p50 MAE, RMSE, and bias. It also reports empirical p10-to-p90 coverage and mean band width:

$$C_{10,90} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{q_{0.10,i} \le y_i \le q_{0.90,i}\}, \tag{14}$$

$$W_{10,90} = \frac{1}{n}\sum_{i=1}^{n}(q_{0.90,i} - q_{0.10,i}). \tag{15}$$

For quantile level α, pinball loss is

$$\rho_\alpha(y,q) = \begin{cases}\alpha(y-q), & y \ge q,\\ (\alpha-1)(y-q), & y < q.\end{cases} \tag{16}$$

The mean score averages Eq. 16 over observations and the three stored quantiles (Koenker and Bassett, 1978; Gneiting and Raftery, 2007). It is not the continuous ranked probability score.

Table 2. Phase 2 station-held-out model metrics.

Model	MAE	RMSE	Bias	n
B0 Persistence	0.350	0.572	0.058	1,555
B1 CropCoefficient	1.532	2.005	0.149	7,923
B2 WeatherRidge	1.514	2.687	-0.098	7,923
M1 OpenETDirect	0.784	1.066	0.154	7,923
M2 OpenETRecal	0.781	1.060	0.005	7,923
M3 OpenETRidge	0.856	1.386	-0.013	7,923

Spatial folds use the target grid cell. Grid 43.50:-116.00 maps to the one-degree tile 43:-116. The fold is the integer value of the full SHA-256 hexadecimal digest for that tile modulo 5. The BOII target maps to fold 2.

The evaluator crosses 20 leads and four seasons: December-January-February (DJF), March-April-May (MAM), June-July-August (JJA), and September-October-November (SON). Five folds give 400 cells. Each cell needs at least 30 paired targets, so the minimum support count is 12,000 cell observations. The paired ETo interval uses 1,000 cluster-bootstrap replicates with seed 20260731. A cluster is one local issue date and target station. An interval requires at least two clusters.

5 Results

5.1 Preregistered OpenET value result

Figure 2 and Table 2 report the committed station-held-out result. The common fitted-model sample is $n = 7,923$ station-days from 85 stations. The preregistered H2 comparison uses M3, not M2.

M3 has MAE 0.856 mm day^{-1}. B2 has MAE 1.514 mm day^{-1}. The paired improvement is 0.658 mm day^{-1}, or 43.4%. The station-blocked 95% interval uses 2,000 replicates. It is [0.399, 0.911] mm day^{-1} and excludes zero under the frozen protocol.

M2 has the lowest point MAE, 0.781 mm day^{-1}. Its direct point reduction from B2 is 48.4%, computed on $n = 7,923$ common rows. This is a descriptive contrast. The result record contains no paired M2 interval. We do not attach the M3 interval to M2.

The Phase 2 result is retrospective. It does not test GEFS, the 20-day horizon, soil-water dynamics, or an irrigation decision.

5.2 Real reference-ETo feasibility diagnostic

The one-issue candidate passes the source count, member count, lead count, quantile order, local-day, station-history, target-time, and grid-distance checks. BOII (the Boise, Idaho AgriMet weather-station identifier) is a retrospective reforecast diagnostic.

Its `source_issue_at` is 2019-07-03 00Z. Its `archive_available_at` is `2026-08-04T18:08:54.243122Z`. The archive time is not public operational availability in 2019. Figure 3 shows its trajectory. The case has one issue, one station, and $n = 20$ daily targets.

The p50 MAE is 1.133 mm day^{-1}. The fixed prior-years station climatology MAE is 0.505 mm day^{-1}. The signed baseline-minus-forecast MAE difference is -0.628 mm day^{-1}, so forecast MAE is higher. The p50 RMSE is 1.295 mm day^{-1}, and the bias is 1.133 mm day^{-1}.

Empirical p10-to-p90 coverage is 0.25, and the nominal p10-to-p90 coverage target is 0.80. The mean p10-to-p90 band width is 1.453 mm day^{-1}. The mean three-quantile pinball loss is 0.387 mm day^{-1}.

These values have no sampling interval. The case has only 1 bootstrap cluster. It also has one target per occupied lead cell, below the frozen minimum of 30. The result is a negative feasibility diagnostic, not an estimate of general forecast skill.

5.3 Spatial artifact and support state

Figure 4 shows the real common 0.5-degree GEFS grid-point subset artifact. Each panel contains 195 weather-grid points. It does not show field boundaries. The gold marker identifies BOII and its mapped grid point.

Figure 5 makes the incomplete state explicit. The real archive supplies one target in each JJA, fold-2 lead cell. This is 20 cell observations. The frozen gate needs 12,000 observations when every one of the 400 cells reaches its minimum. The present archive therefore supports no full-hindcast claim.

6 Discussion

6.1 What the Phase 2 result establishes

The completed result supports one narrow conclusion. Under the frozen station-held-out split, M3 OpenETRidge reduces retrospective daily actual-ET MAE against B2 WeatherRidge. The paired interval accounts for station clustering. It does not account for a new geographic domain, a new label system, or a future weather issue. The comparison therefore supports OpenET value within this benchmark. It does not validate the outlook.

M2 creates an important reporting test. Its point MAE is lower than M3. A reader could reasonably call M2 the descriptive winner. The preregistered H2 arm is still M3 because the runner fixed that model. A paper must report both facts. Combining M2's point error with M3's interval would create a result that no execution produced.

The persistence row has a much lower MAE than every fitted model. That value uses only 1,555 consecutive-day pairs and consumes the previous observed label. It is not available for a lead-1 forecast without an observed prior day. We therefore treat it as an oracle-like diagnostic, not the H2 baseline.

6.2 What the feasibility case establishes

The reference-ET_o case shows that the data path can produce an auditable artifact. The path starts with an archived issue and ends with a later, independent station target. It preserves local-day semantics, members, quantiles, source receipts, and checksums. This is a systems result.

The diagnostic forecast is worse than the simple climatology for the one case. Its median forecast also has a positive bias of 1.133 mm day^{-1}. The empirical p10-p90 band covers 25% of the 20 targets. These numbers warn against promotion. They do not identify a stable lead effect, season effect, or statewide effect. One issue and one station cannot estimate those effects.

The incomplete state is scientifically useful. It prevents source feasibility from becoming forecast validation. It also

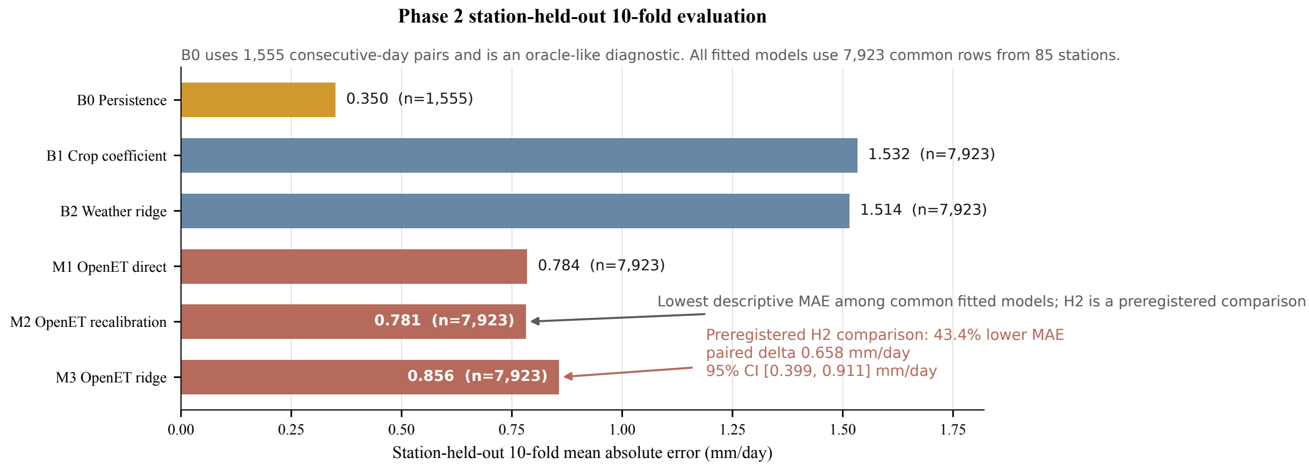

Figure 2. Station-held-out actual-ET MAE. B2 and M3 form the preregistered H2 comparison. Their common sample contains $n = 7,923$ rows from 85 stations. M2 has the lowest descriptive point MAE, but it does not inherit the M3 interval. B0 uses $n = 1,555$ consecutive-day pairs and is excluded from H2.

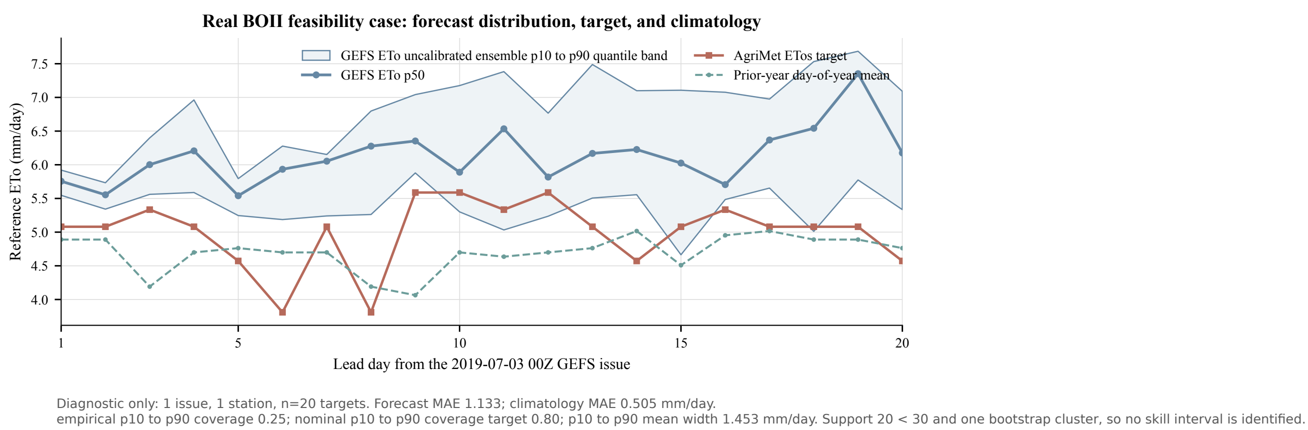

Figure 3. One real BOII feasibility case. The GEFS member distribution gives p10, p50, and p90 reference ET_o for leads 1 through 20. The independent target is AgriMet ETos. The baseline is the prior-year station and day-of-year mean. The scope is one issue, one station, and $n = 20$ targets. It is diagnostic only.

preserves the negative diagnostic instead of deleting it. A full result must include every frozen support cell, even when the candidate does not improve on climatology.

6.3 Why target separation matters

Reference ET_o is a standardized reference evapotranspiration rate. Actual ET is a realized surface flux. An actual-ET outlook also needs crop state, root-zone water, and management assumptions. A crop coefficient can convert ET_o to potential crop ET under named assumptions. It cannot by itself create unconditional actual ET.

This boundary affects model assessment. A retrospective satellite feature can help predict historical actual ET. That evidence does not make the feature available at a future issue. An AgriMet ETos target can assess future reference ET_o. That evidence does not assess field-scale actual ET. The ontology in Table 1 keeps these claims separate by construction.

7 Limitations

The paper has six main limitations.

1. The full 365-issue outcome archive is absent. Reference-ET_o skill remains **evaluation pending**.
2. The current AgriMet registry is not a historical movement ledger. Historical page evidence covers 19 stations.
3. The BOII location evidence brackets the case date. It does not prove an unchanged coordinate for every date from 2013 through 2019.
4. The spatial artifact contains weather-grid points. It does not contain field boundaries, crop labels, or area weights.
5. The Phase 2 comparison uses public flux labels and historical OpenET. It does not measure future forecast performance.
6. The one-case ETo metrics have $n = 20$ and one cluster. Their uncertainty is not estimable under the frozen bootstrap.

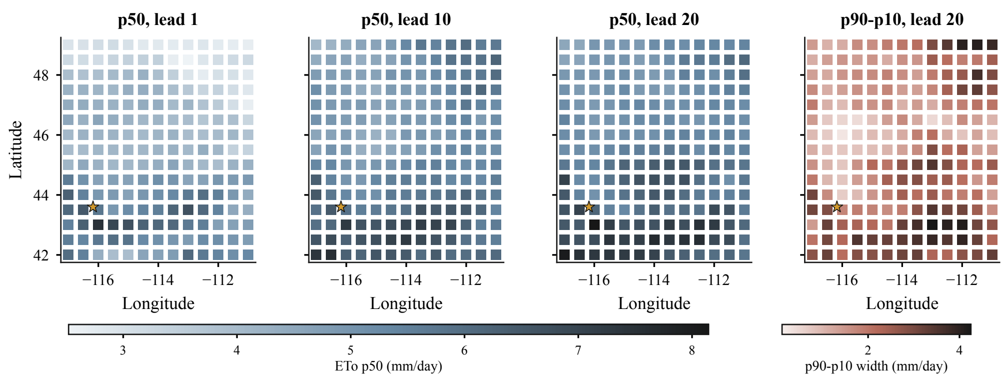

Figure 4. Reference-ET_o fields for the real 2019-07-03 candidate on the common 0.5-degree GEFS grid-point subset. The first three panels show p50 at leads 1, 10, and 20. The last panel shows p90 minus p10 at lead 20. Each panel contains n = 195 grid points. No interpolation, field geometry, or area-weighted Idaho estimate is inferred.

8 Conclusions

MLET separates retrospective actual ET from prospective reference ET_o. The Phase 2 record supports a preregistered M3 versus B2 actual-ET result. M3 reduces MAE by 43.4%, with a paired 95% interval of [0.399, 0.911] mm day^{-1}. M2 has a lower descriptive point MAE, but it has no recorded paired interval.

The reference-ET_o path produces a real, checksum-bound candidate and an independent target. Its one-case diagnostic is worse than climatology and has insufficient support. The paper therefore makes no reference-ET_o skill or irrigation claim. Promotion requires the complete frozen hindcast and a separate release review.

Data and code availability

The repository stores the Phase 2 result, its independent reproduction receipt, the GEFS feasibility receipts, the AgriMet target artifacts, and the schema-v4 feasibility archive. The manuscript figures read those artifacts directly. The scripts `scripts/build_arxiv_claims.py` and `scripts/build_arxiv_figures.py` regenerate the numerical macros and figures. The complete raw GEFS plan is approximately 3.026 TB by a single-issue multiplication estimate. That estimate uses 365 planned issues and one 8,289,206,079-byte inventory. It has no sampling uncertainty and is not a measured full transfer.

Reproducibility statement

The Phase 2 protocol uses seed 20260713 and 2,000 station-blocked replicates. The ETo protocol uses seed 20260731 and 1,000 issue-station cluster replicates. Generated claims fail when B2 minus M3 does not equal the H2 delta. They also fail when M2 is not the descriptive minimum or when the ETo evidence scope changes. Appendix B records the exact artifact digests used here.

A Implementation map

Table 3 maps each manuscript operation to the local implementation. This map is part of the claim audit. It is not a substitute for the source code.

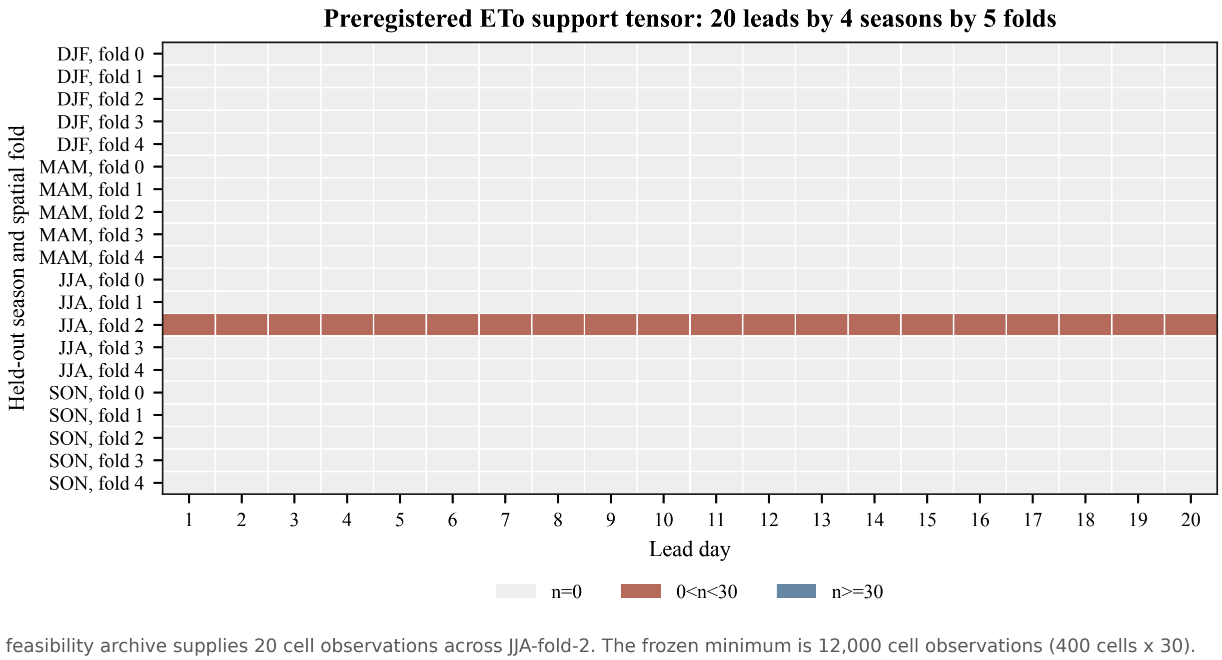

Figure 5. Frozen reference-ET_o support tensor. The real feasibility archive occupies 20 cells with one target each. No cell reaches the $n \geq 30$ threshold. The complete protocol requires all 400 lead-season-fold cells.

Table 3. Equation and claim map to the implementation.

Operation	Local implementation	Enforced invariant
Phase 2 join	`src/mlet/build_dataset.py`	Exact station and date key; `ET_corr` label
Model definitions	`src/mlet/baselines.py`	Training-only fit; ridge penalty 1.0
Blocked H2 interval	`src/mlet/evaluate.py`	Station blocks; 2,000 replicates
Phase 2 runner	`src/mlet/experiments/phase2_openet_value.py`	H2 arm is M3 versus best B1 or B2
GEFS decoder script	`scripts/decode_gefs_reforecast.py`	Re-hashes one verified issue before artifact output
GEFS batch decoder	`src/mlet/sources/gefs_reforecast_batch.py`	Complete member and component plan
GRIB parser	`src/mlet/sources/gefs_grib.py`	Fixed short names and grid identity
Daily GEFS artifact	`src/mlet/sources/gefs_reforecast.py`	Local day; 10 m wind and component identity
Reference ET_o	`src/mlet/outlook/eto.py`	ASCE short reference; p10, p50, p90
Candidate contract	`src/mlet/outlook/eto_build.py`	20 leads; research status; no extra layers
AgriMet target	`src/mlet/sources/agrimet.py`	Independent ETos; exact 25.4 conversion
Baseline builder	`scripts/build_eto_target_index.py`	Prior years only for station and day of year
Outcome evaluator	`src/mlet/outlook/eto_hindcast.py`	400 cells; fail-closed completion state

B Artifact ledger

The following digests bind the plotted evidence. Paths are relative to the repository root.

Phase 2 result

`docs/results/phase2_openet_value.json`

SHA-256: `2a2cece4a4febaa6105d5182ae01e161b98db7c18bb0ae499310fe160cd5eee3`

ETo feasibility evidence

`data/outlook/eto_feasibility_archive/evidence.json`

SHA-256: `3e1f022ea7e0e6f280f9854adae7080f4c308b57da70a578ce9d6c10ae95b075`

ETo evaluation

Deterministic evaluation SHA-256: `b2ab81b0463d8a01285c05604c32e26dd43335e35052584674ce3413014729c7`

BOII outlook

`data/outlook/eto_feasibility_archive/cases/`
`issue-20190703-station-BOII-season-JJA-fold-2/outlook.json`

SHA-256: `c0b51331b20f1d15f51de8f0b6a9c4efd8d2a9801fa617686a933b4d07e45163`

BOII target

`data/outlook/eto_feasibility_archive/cases/`
`issue-20190703-station-BOII-season-JJA-fold-2/target.json`

SHA-256: `1d5570ab9baf7a4bb978be2069dbff8c89b0f88546442884556f37eb481eb43f`

C Claim ledger

Table 4. Publication claims and their current states.

State	Claim	Evidence boundary
Supported	M3 reduces retrospective actual-ET MAE against B2 by 43.4%.	$n = 7,923$, 85 stations, 2,000 station-blocked replicates, 95% interval [0.399, 0.911] $\mathrm{mm\,day^{-1}}$.
Supported	M2 has the lowest descriptive MAE at 0.781 $\mathrm{mm\,day^{-1}}$.	$n = 7,923$ common rows; no paired M2 interval in the record.
Diagnostic	One BOII ETo case has MAE 1.133 $\mathrm{mm\,day^{-1}}$.	One issue, one station, $n = 20$, one bootstrap cluster.
Pending	The 20-day reference-ET_o outlook improves on climatology.	Full 365-issue archive, all 400 cells, and release review required.
Out of scope	The artifact predicts field-scale actual ET or irrigation need.	No field target, crop state, soil-water target, or irrigation outcome exists.

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